

MEMORANDUM

Semester Test 2

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Electricity and Electronics EBN 122

5 October 2019

Test Information

Maximum marks:	45	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		16	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form, etc.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. de Villiers, Prof J.P. Jacobs, Dr X. Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the next 90 minutes of the test. (0-90 min)

Do step 2 within the next 10 minutes. (90-100 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

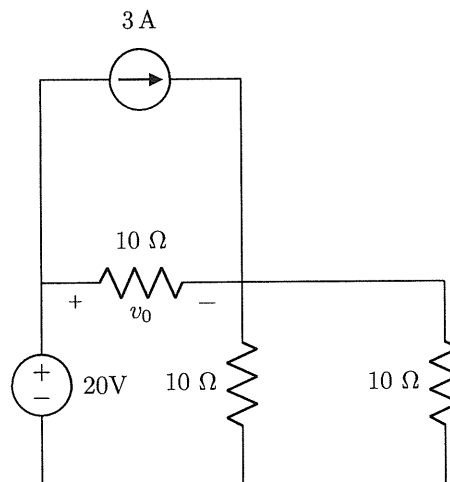
STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122S02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

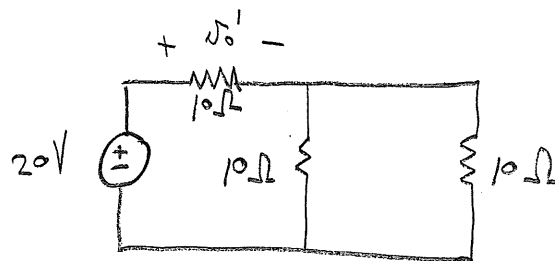
Section 1

Questions 1 to 2 [4]

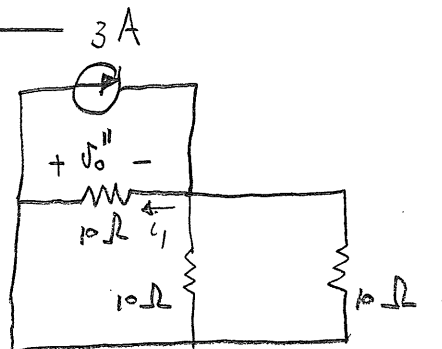
Find the individual contributions of each of the 20 V and 3 A sources to the voltage v_o .

01 Contribution of 20 V source [2]

02 Contribution of 3 A source [2]

20 V source:

$$v_o' = 20 \frac{10}{10 + 10 \parallel 10} = 13.33 \text{ V}$$

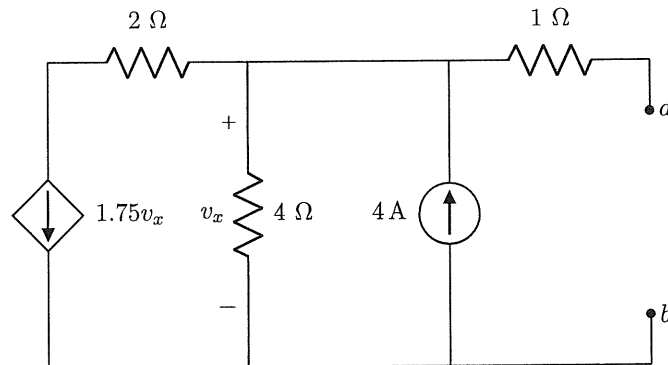
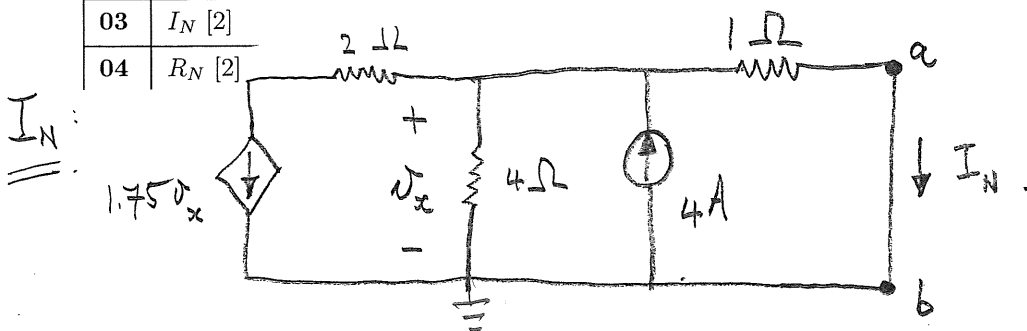
3 A source:

$$i_1 = 3 \frac{10 \parallel 10}{10 + 10 \parallel 10} = 1 \text{ A}$$

$$v_o'' = -i_1 \cdot 10 = -10 \text{ V}$$

Questions 3 to 4 [4]

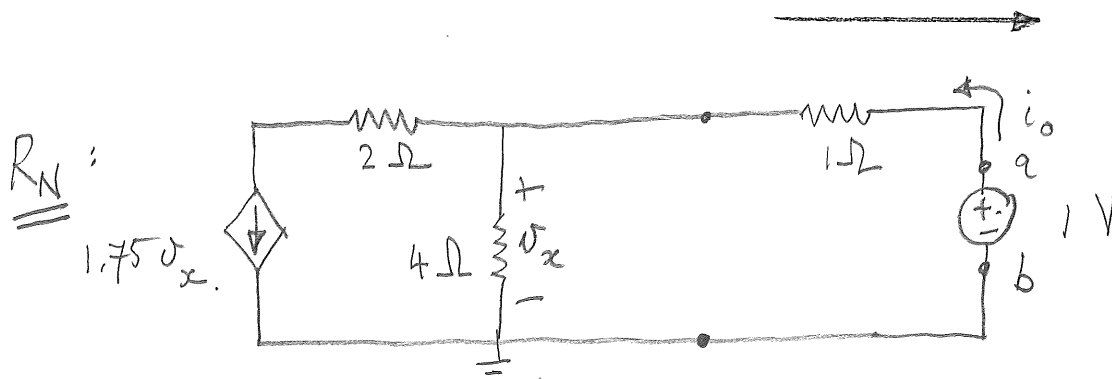
Find the Norton equivalent current I_N , and Norton equivalent resistance R_N , with reference to terminals $a - b$.

03 I_N [2]04 R_N [2]

$$\text{KCL: } 1.75v_x + \frac{v_x}{4} + \frac{v_x}{1} = 4$$

$$7v_x + v_x + 4v_x = 16 \quad \therefore v_x = \frac{16}{12} = 1.333 \text{ V}$$

$$I_N = \frac{v_x}{1} = 1.333 \text{ A}$$



$$\text{KCL: } 1.75v_x + \frac{v_x}{4} + \frac{v_x - 1}{1} = 0$$

$$7v_x + v_x + 4v_x = 4$$

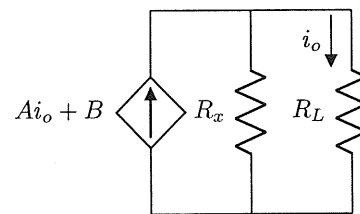
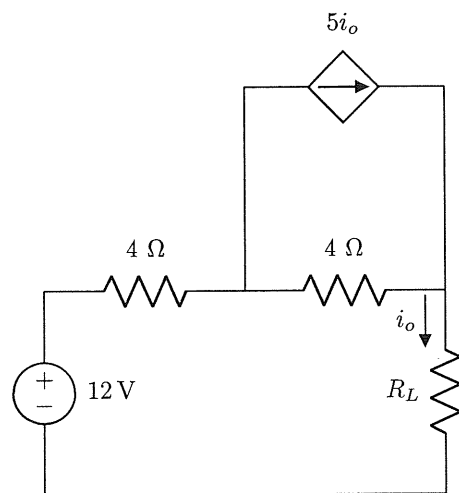
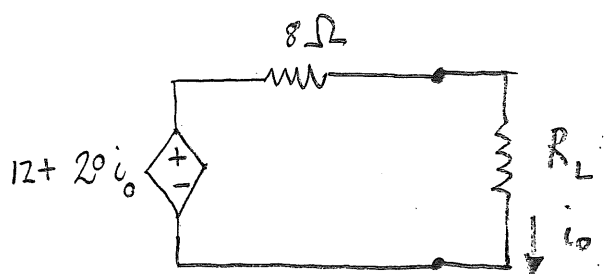
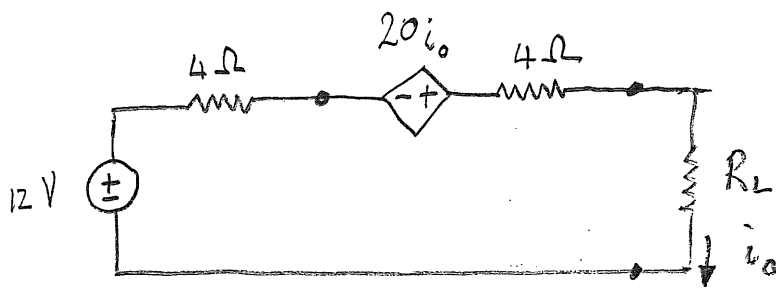
$$\therefore v_x = \frac{4}{12} = 0.333 \text{ V}$$

$$i_o = \frac{1 - v_x}{1} = 0.667 \text{ A}$$

$$R_N = \frac{1}{i_o} = 1.5 \Omega$$

Questions 5 to 7 [4]

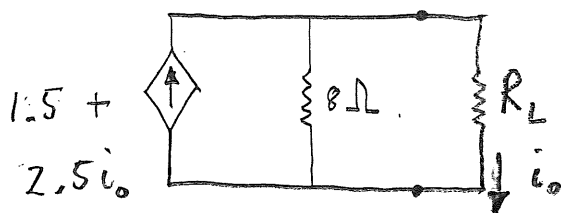
Simplify the circuit on the left by means of source transformations so that it looks like the circuit on the right. Determine R_x , and the coefficients A and B .

05 R_x [2]06 A [1]07 B [1]

$$R_x = 8\Omega$$

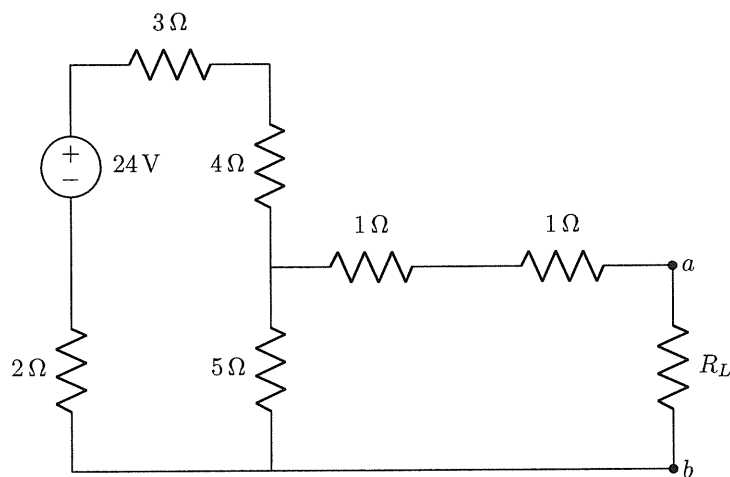
$$A = 2.5$$

$$B = 1.5$$



Questions 8 to 10 [5]

Consider the circuit below.



08	Find the Thevenin equivalent voltage V_{Th} seen by the load R_L . [2]
09	Find the value of R_L so that maximum power is transferred to R_L . [2]
10	Suppose the value of R_L is increased to twice the value that was obtained in the previous question. Would V_{Th} change? Write 1 for yes, or 2 for no. [1]

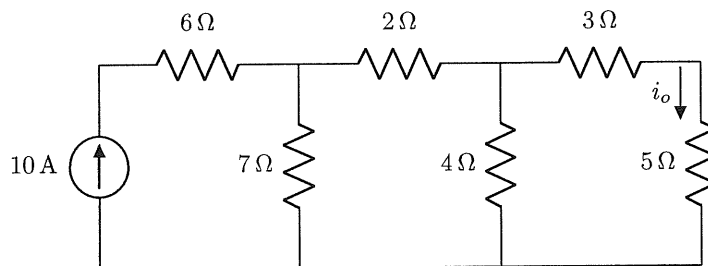
$$\underline{V_{Th}}: \quad V_{Th} = 24 \cdot \frac{5}{2 + 3 + 4 + 5} = \underline{8.571 \text{ V}}$$

$$\begin{aligned} \underline{R_{Th}}: \quad R_{Th} &= (1 + 1) + 5 \parallel (2 + 3 + 4) \\ &= 2 + \frac{5 \cdot 9}{5 + 9} \\ &= \underline{5.214 \Omega = R_L} \end{aligned}$$

$$No. \equiv 2$$

Question 11 [2]

It is known that $i_o = 2$ A for the circuit below. Find the new value of i_o if the value of the current source is increased to 11 A.



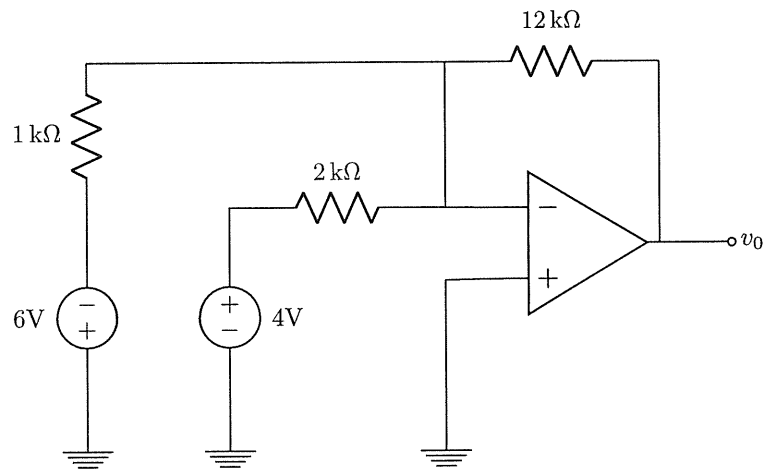
11 i_o [2]

$$i_o = \frac{11}{10} \cdot 2 = 2.2 \text{ A.}$$

Total Section 1: [19]

Section 2

Question 12 [2]

Assume that the op-amp is ideal. Find v_o .

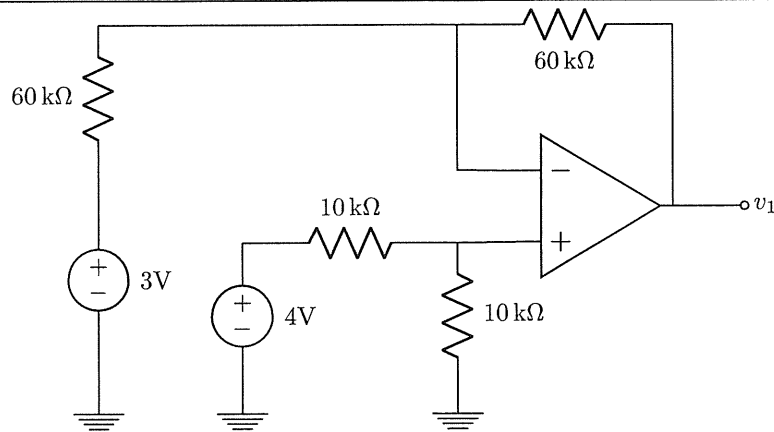
12

 v_o [2]

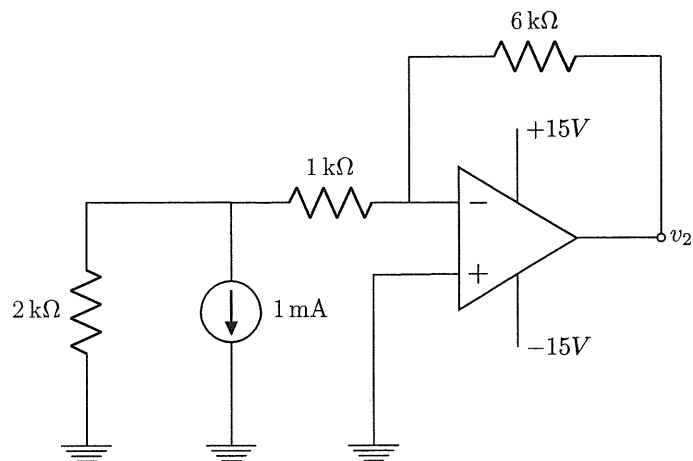
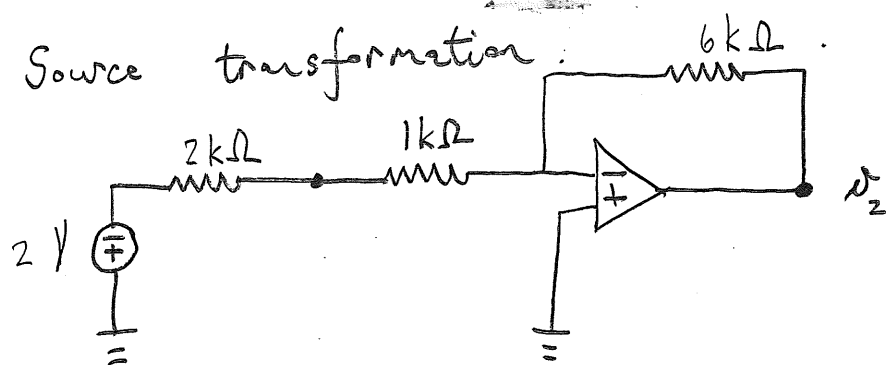
Summing amplifiers:

$$v_o = - \left(\frac{12}{1} (-6) + \frac{12}{2} (4) \right)$$
$$= 48V$$

Question 13 [2]

Assume that the op-amp is ideal. Find v_1 .13 v_1 [2]

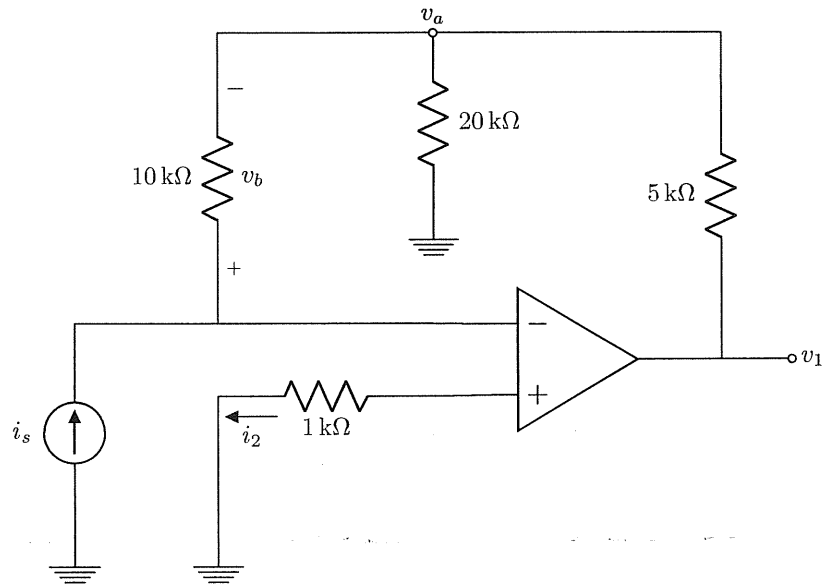
Difference amplifier $\rightarrow$ subtractor since
 $R_2 = R_1$ and $R_3 = R_4$
 $v_1 = 4 - 3 = 1 \text{ V}$

Question 14 [2]Assume that the op-amp is ideal. Find v_2 .14 v_2 [2]

$$v_2 = - \frac{6}{2+1} (-2) = 4 \text{ V.}$$

Questions 15 to 18 [6]

Assume that the op-amp is ideal, and that $v_a = -40$ V. Find i_s , i_2 , v_1 and v_b .

15 i_s [2]16 i_2 [1]17 v_1 [2]18 v_b [1]

$$\underline{i_s}: \quad i_s = \frac{0 - v_a}{10000} = \frac{0 - (-40)}{10000} = 4 \text{ mA} \rightarrow$$

$$\underline{i_2}: \quad i_2 = 0 \text{ A} \rightarrow$$

$$\underline{v_1}: \quad \frac{v_a - 0}{10} + \frac{v_a}{20} + \frac{v_a - v_1}{5} = 0$$

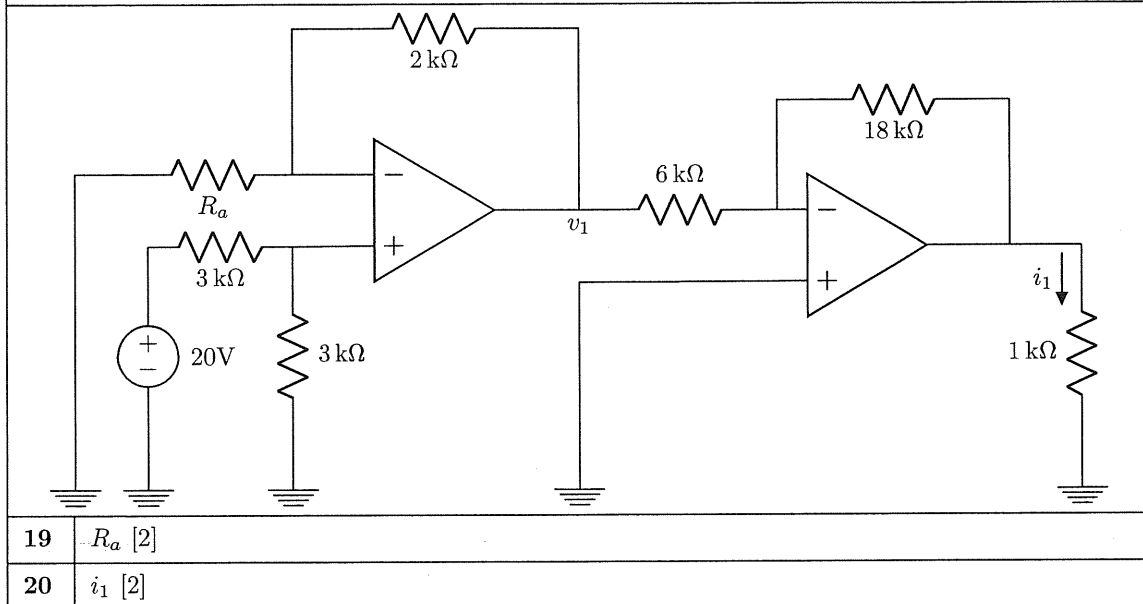
$$2v_a + v_a + 4v_a = 4v_1$$

$$v_1 = \frac{1}{4} (2 + 1 + 4) (-40) = -70 \text{ V} \rightarrow$$

$$\underline{v_b}: \quad v_b = -v_a = +40 \text{ V} \rightarrow$$

Questions 19 to 20 [4]

Two ideal op-amps are connected as shown. It is known that $v_1 = 15$ V. Find R_a and i_1 .



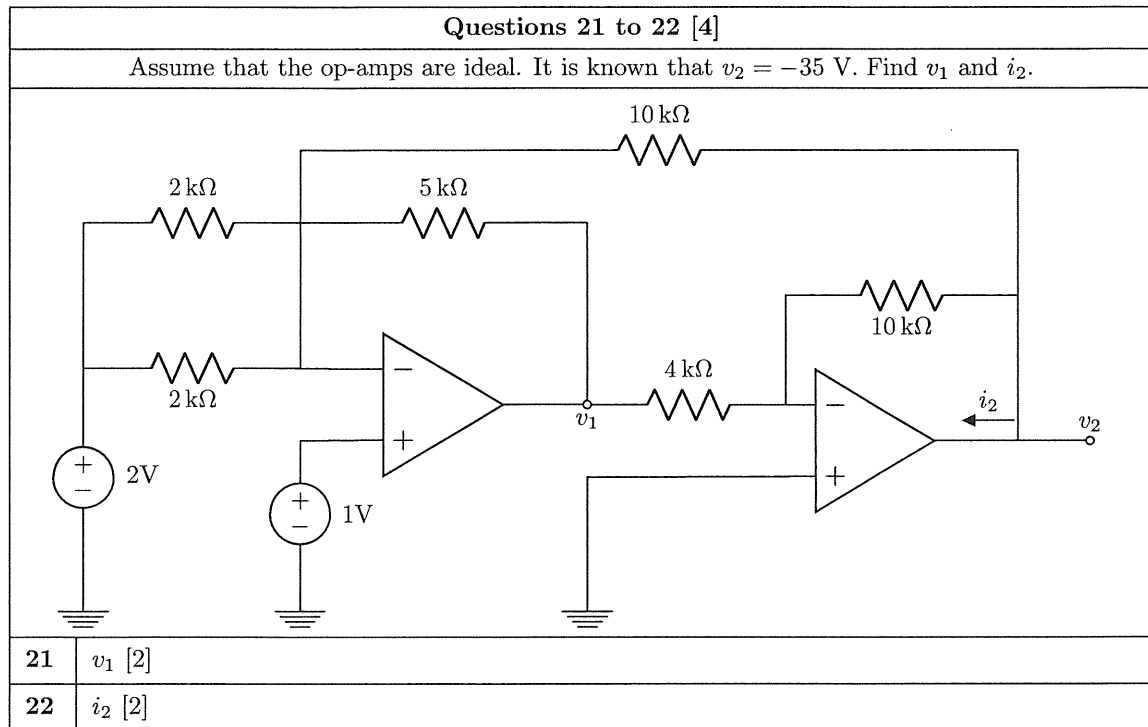
$$\underline{R_a} : \quad v_1 = \left(1 + \frac{2000}{R_a}\right) \left(20 \cdot \frac{3}{3+3}\right)$$

$$15 = 10 + \frac{2000}{R_a} 10$$

$$5 = \frac{2000}{R_a} 10 \quad \therefore \underline{R_a = 4 \text{ k}\Omega}$$

$$\underline{i_1} : \quad v_2 = -\frac{18}{6} v_1 = -3 \cdot 15 = -45 \text{ V}$$

$$i_1 = \frac{v_2}{1000} = \underline{-45 \text{ mA}}$$



v_1 : KCL at - terminal of left op-amp:

$$2 \parallel 2 = 1 \text{ k}\Omega$$

$$\frac{1-2}{1} + \frac{1-v_1}{5} + \frac{1-v_2}{10} = 0$$

$$-10 + 2 - 2v_1 + 1 - (-35) = 0$$

$$-2v_1 = -28 \therefore \underline{v_1 = 14 \text{ V}}$$

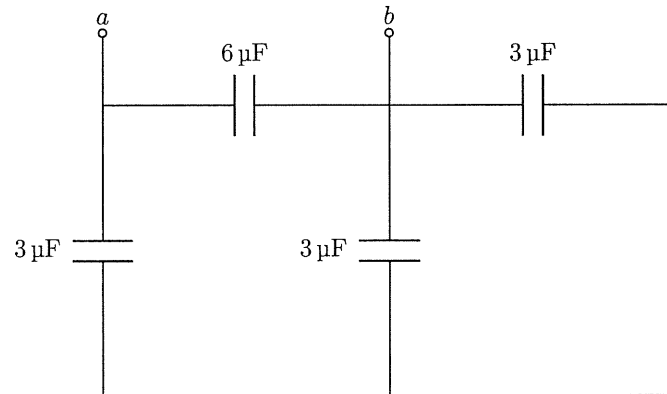
i_2 :
$$i_2 + \frac{v_2 - 0}{10\,000} + \frac{v_2 - 1}{10\,000} = 0$$

$$10\,000 i_2 + v_2 + v_2 - 1 = 0$$

$$\therefore i_2 = \frac{1}{10\,000} (-2v_2 + 1)$$

$$= 7.1 \text{ mA}$$

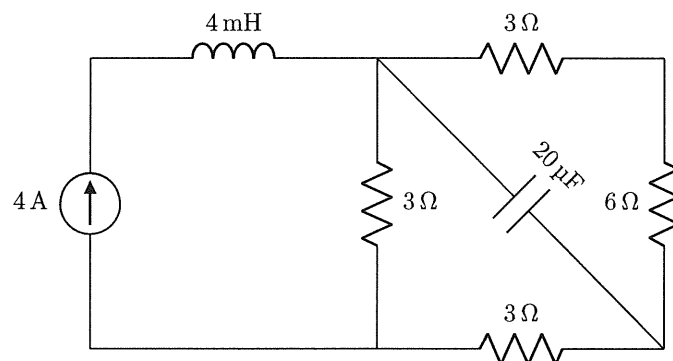
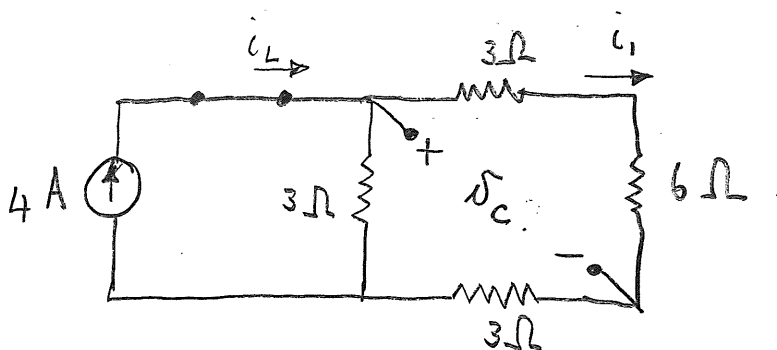
Total Section 2: [20]

Section 3**Question 23 [2]**Find the equivalent capacitance C_{eq} with respect to terminals $a - b$.**23** C_{eq} [2]

$$C_{eq} = 6 \parallel (3 \text{ in series with } 3 \parallel 3)$$
$$= 6 + \left(\frac{3 \cdot 6}{3 + 6} \right) = 8 \mu F$$

Questions 24 to 25 [4]

Under DC conditions, calculate w_C , the energy stored in the capacitor; and w_L , the energy stored in the inductor.

24 w_C [2]25 w_L [2]

$$i_L = 4A \quad \therefore w_L = \frac{1}{2} L i_L^2 = \frac{1}{2} (4 \times 10^{-3}) 4^2 = 32 \text{ mJ}$$

$$i_1 = 4 \frac{3}{3 + (3+6+3)} = 0.8 \text{ A}$$

$$v_C = (3+6) i_1 = 7.2 \text{ V}$$

$$w_C = \frac{1}{2} C v_C^2 = \frac{1}{2} (20 \times 10^{-6}) (7.2)^2 = 0.518 \text{ mJ}$$

Total Section 3: [6]

Grand Total : [45]